Week 5 — In-Class Exercise

From response to uplift to a rule a brand team can run

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Groups of 3–4 · 20 minutes + 5-minute debrief · one sheet per group

Materials: this sheet, a calculator (or one notebook cell), tonight's slides 6, 18, 24 and 25

Part A · Uplift by hand (7 minutes)

Eight GLUCOREN prescribers from the exploration arm. For each, the mean NBRx over the following two weeks when the randomized action was none, a rep call, or an approved email (each over about six randomized weeks), the segment, the touches already delivered in the trailing four weeks, and email consent. Conventions: $1,200 per incremental NBRx; $180 per call; $2 per email.

HCP	Segment	Touches (4 wks)	Email consent	NBRx | none	NBRx | call	NBRx | email
Dr. Adler	Core	1	yes	4.60	4.70	4.75
Dr. Bakshi	Movers	1	yes	2.00	2.90	2.30
Dr. Costa	Entrenched	2	yes	0.50	0.30	0.45
Dr. Dunn	Long tail	0	yes	0.25	0.30	0.28
Dr. Eng	Access-constrained	1	yes	1.40	1.45	1.75
Dr. Farah	Core	4	yes	5.00	4.70	4.90
Dr. Goyal	Movers	0	no	1.60	2.30	1.60
Dr. Hale	Core	2	yes	6.20	6.30	6.25

1. For each HCP compute: response to the call (NBRx | call), uplift of the call (NBRx | call − NBRx | none), net value of the call (uplift × $1,200 − $180), and the same for the email.
2. Label each HCP: sure thing, persuadable, sleeping dog, lost cause — or 'ineligible', and say which rule.
3. Rank the eight by response to the call. Which two does a response model rank first? Which would the NBA engine actually touch this week, with which action? Where does the money leak?

HCP	Response (call)	Uplift (call)	Net $ (call)	Uplift (email)	Net $ (email)	Kind / rule	Engine's action
Dr. Adler							
Dr. Bakshi							
Dr. Costa							
Dr. Dunn							
Dr. Eng							
Dr. Farah							
Dr. Goyal							
Dr. Hale							

Part B · Write the rules (8 minutes)

Your five segments from Week 4 (Core, Movers, Access-constrained, Entrenched, Long tail), tonight's saturation curve (slide 24) and the affinity matrix (slide 18). For each segment write: the default action for a

typical week (check it against the affinity matrix), the frequency cap, one suppression rule, and one trigger that changes the action. Then answer the two questions below the table.

Segment	Default action	Cap (touches / 4 wks)	One suppression rule	One trigger that changes the action
Core				
Movers				
Access-constrained				
Entrenched				
Long tail				

4. Which segment should the engine leave alone, and what would make you change your mind?
5. The field director says: 'your engine cut rep calls from 955 to 281 a week.' Write the two-sentence answer — one sentence with a number from tonight, one with what you would offer to measure.

Part C · Pick the evaluation (5 minutes)

For each scenario: offline policy evaluation (IPS / doubly robust), a global holdout, an A/B test, a geo test, or 'this cannot be evaluated causally'. One sentence of justification each.

#	Scenario	Method and why
1	The brand wants to know what the NBA engine would earn vs. the legacy engine before any field change. The log has 25% randomized HCP-weeks and the legacy rules were logged as probabilities.	
2	The engine has been live for two quarters with a 10% global holdout. The CFO asks what the program earned in total.	
3	Analytics has a new uplift model (X-learner) and wants to replace the old one (T-learner) in production.	
4	The digital agency buys placements by DMA and cannot log a propensity per HCP.	
5	The field director refused any randomization; the only data is the CRM log of who was called and when.	
6	Someone proposes a policy that calls every HCP every week and asks for its OPE value.	